Worksheet 36 · 3D Trigonometry IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Identify right-angled triangles in 3D shapes • Find lengths in 3D shapes using Pythagoras' theorem • Find the angle between a line and a plane		
Grade: / 51	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Questions *3D Trigonometry*

Question 1 (4 marks)

The diagram shows a cuboid $ABCDEFGH$.

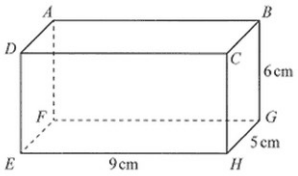


Diagram **NOT** accurately drawn

$EH = 9\,\text{cm}$, $HG = 5\,\text{cm}$ and $GB = 6\,\text{cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

°

Question 2 (4 marks)

Here is a cube $ABCDEFGH$.

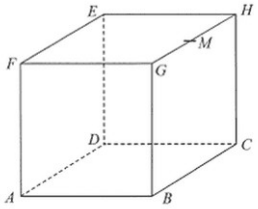


Diagram **NOT** accurately drawn

M is the midpoint of the edge GH.

Find the size of the angle between the line MA and the plane $ABCD$.
Give your answer correct to 1 decimal place.

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Question 3 (3 marks)

The diagram shows a triangular prism.

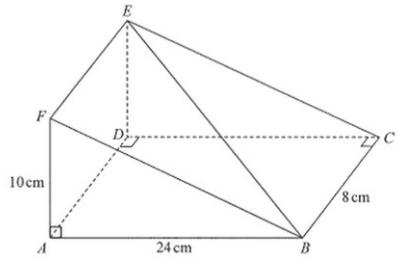


Diagram **NOT** accurately drawn

$AF = 10\,\text{cm}$, $AB = 24\,\text{cm}$ and $BC = 8\,\text{cm}$.
Angle FAB = angle ADC = angle $BCD = 90°$

Work out the size of the angle between the line BE and the plane $ABCD$.
Give your answer correct to 1 decimal place.

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Question 4 (6 marks)

The diagram shows the prism $ABCDEF$ with cross section triangle ABC.

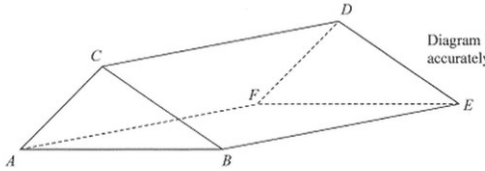


Diagram **NOT** accurately drawn

Angle $BEC = 40°$ and angle ACB is obtuse.
$AC = 6\,\text{cm}$ and $CE = 13\,\text{cm}$

The area of triangle ABC is $22\,\text{cm}^2$

Calculate the length of AB.
Give your answer correct to one decimal place.

..... cm

Question 5 (4 marks)

The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

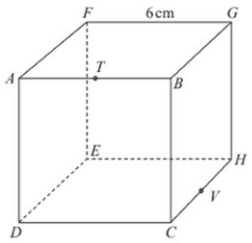


Diagram **NOT** accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.
Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

Question 6 (7 marks)

The diagram shows a solid prism $ABCDEFGH$.

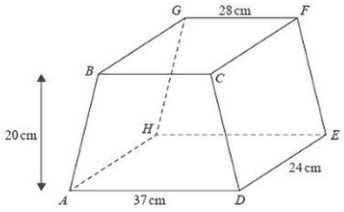


Diagram **NOT** accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC, is a cross section of the prism.
The base $ADEH$ of the prism is a horizontal plane.
$ADEH$ and $BCFG$ are rectangles.
The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$AD = 37$ cm $GF = 28$ cm $DE = 24$ cm

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm²
(4)

(b) Calculate the size of the angle between AF and the plane $ADEH$.
Give your answer correct to one decimal place.

.....
(3)

Question 7 (4 marks)

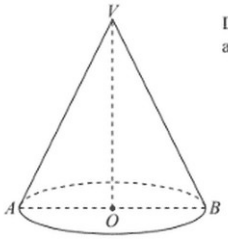


Diagram **NOT** accurately drawn

The diagram shows a solid cone.
The base of the cone is a horizontal circle, centre O, with radius 4.5 cm.
AB is a diameter of the base and OV is the vertical height of the cone.
The curved surface area of the cone is 130 cm²

Calculate the size of the angle AVB.
Give your answer correct to 1 decimal place.

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Question 8 (4 marks)

The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

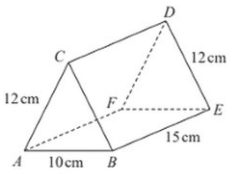


Diagram **NOT** accurately drawn

$AC = BC = FD = ED = 12$ cm $AB = 10$ cm $BE = 15$ cm

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

.....°

Question 9 (4 marks)

ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B.

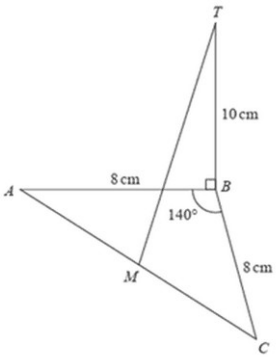


Diagram **NOT** accurately drawn

Angle $ABC = 140°$
$AB = BC = 8$ cm
$TB = 10$ cm
M is the midpoint of AC.

Calculate the size of the angle between MT and the horizontal plane ABC.
Give your answer correct to one decimal place.

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Question 10 (6 marks)

The diagram shows a solid pyramid $ABCDE$ with a horizontal base.

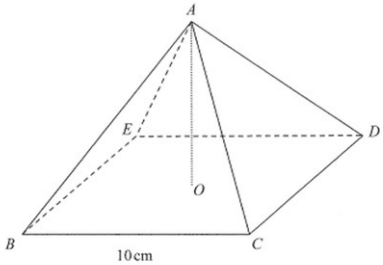


Diagram **NOT** accurately drawn

The base, $BCDE$, of the pyramid is a square of side 10 cm.

The vertex A of the pyramid is vertically above the centre O of the base so that
$AB = AC = AD = AE$

The **total** surface area of the pyramid is 360 cm²

Work out the size of the angle between AC and the base $BCDE$.
Give your answer correct to 3 significant figures.

Question 11 (5 marks)

The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

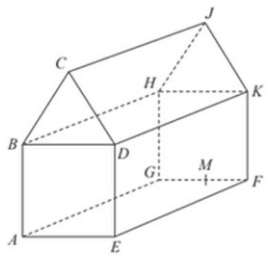


Diagram **NOT** accurately drawn

$ABCDE$ is a cross section of the prism where
$ABDE$ is a square
BCD is an equilateral triangle

$EF = 2 \times AE$
M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y°$

Given that $\tan y° = T$

find the value of T, giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T =$ .....